

# AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$ -SYMMETRIC TRIPLE-OCTONION PRODUCT

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**ABSTRACT.** We show that the Leech lattice  $\Lambda$  admits the structure of an order in a 24-dimensional real algebra. The product is a  $\mathbb{Z}_3$ -symmetric triple-octonion product on  $\mathbb{R}^{24}$ , assembled from nine octonion-multiplication blocks under cyclic  $\mathbb{Z}_3$  permutation of three octonion factors. Realising this product concretely on Wilson’s sublattice description of  $\Lambda \subset \mathbb{R}^{24}$  requires a particular Coxeter representation of the octonions, related to the standard Fano-plane labelling by a single transposition  $\sigma$  of two imaginary basis indices. The chain Kirmse  $\rightarrow$  Coxeter  $\rightarrow \sigma$  — each step a transposition of two basis indices — successively aligns the octonion multiplication with the lattice: Coxeter’s transposition turns Kirmse’s  $E_8$  candidate into an order in the octonions, and  $\sigma$  aligns the triple-octonion product so that it closes on  $\Lambda$ .

This result was developed through a systematic human–AI collaboration; the complete research record is publicly available [23].

## 1. INTRODUCTION

The Leech lattice  $\Lambda$  is the unique even unimodular lattice of rank 24 with no vectors of squared norm 2 [4]. Its minimal shell  $\text{Min}(\Lambda)$  consists of 196,560 vectors of squared norm 8, and its automorphism group is the Conway group  $\text{Co}_0$ , a double cover of the sporadic simple group  $\text{Co}_1$ .

In rank 8, the  $E_8$  lattice  $L$  carries an octonionic multiplication that makes it a maximal order in the real octonion division algebra. There are seven such orders, one for each of the seven maximal orders of the integral octonions [6, 5]. Wilson [22] showed that  $\Lambda$  embeds naturally in  $L^3$  via three sublattice conditions, raising the question: does  $\Lambda$  inherit an algebraic structure from the octonion product?

This question has been studied from several angles. Dixon [8] constructed  $\Lambda$  using an octonionic XY-product. Baez and Egan [2] explored Jordan-algebraic structures on  $\Lambda$  within the exceptional Jordan algebra  $\mathfrak{h}_3(\mathbb{O})$ , finding closure under a doubled Jordan product.

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*Date:* 28 April 2026.

*2020 Mathematics Subject Classification.* 11H06, 17A75, 11H56.

*Key words and phrases.* Leech lattice, octonions, non-associative order, integral algebra.

The  $\mathbb{Z}_3$ -symmetric triple-octonion product on  $L^3$  — using the same octonion multiplication in all three blocks with  $\mathbb{Z}_3$ -symmetric cross-block routing — satisfies two of Wilson’s three sublattice conditions automatically, but closure on the third requires a block-sum sublattice  $Ls$  to be preserved under octonion multiplication, and that condition is sensitive to which Coxeter representation of the octonions is used in  $\mathbb{R}^{24}$ .

The main result of this paper is that the  $\mathbb{Z}_3$ -symmetric triple-octonion product closes on  $\Lambda$  once the octonions are realised in the Coxeter representation that is aligned with Wilson’s sublattice description of  $\Lambda$ . Relative to the standard Fano-plane labelling we use throughout for the  $E_8$  block, that representation is obtained by a single transposition  $\sigma$  of two imaginary basis indices: the resulting product maps the block-sum sublattice  $Ls$  into  $\sigma(Ls)$ , a different sublattice of  $L$  that is closed under the underlying octonion multiplication.

**Theorem 1.1.** *Let  $\cdot_\sigma$  denote the transposition-twisted octonion multiplication on  $\mathbb{R}^8$  (Definition 3.1), and let  $\star$  denote the associated triple product on  $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$  with  $\mathbb{Z}_3$  cross-block routing (Definition 3.5). Then for all  $u, v \in \Lambda$ , the product  $u \star v$  lies in  $\Lambda$ .*

**Corollary 1.2.**  *$(\Lambda, +, \star)$  is an order in the  $\mathbb{R}$ -algebra  $(\mathbb{R}^{24}, +, \star)$ .*

*Proof.*  $\text{Min}(\Lambda)$  generates  $\Lambda$  over  $\mathbb{Z}$  [4]. The product  $\star$  is bilinear (it is a sum of bilinear octonion products). By Theorem 1.1,  $\star$  maps  $\Lambda \times \Lambda$  into  $\Lambda$ .  $\square$

*Remark 1.3* (Order-closure, not shell-closure). The closure statement  $\Lambda \star \Lambda \subseteq \Lambda$  means that products land somewhere in  $\Lambda$ ; it does not assert that the minimal shell is preserved. Empirically, the product norm on  $\text{Min}(\Lambda) \times \text{Min}(\Lambda)$  takes values in  $\{16, 32, 48, 64, 80, 96, 112, 128\}$  (see Section 5), none of which equals the minimal value  $N = 8$ : products of minimal vectors always land on strictly higher shells. This is consistent with  $(\Lambda, +, \star)$  being an order in the ring-theoretic sense — a subring that is a free  $\mathbb{Z}$ -module of rank equal to the ambient algebra’s dimension — a property weaker than, and independent of, any shell-level closure.

## 2. PRELIMINARIES

**2.1. The octonion algebra.** Let  $\mathbb{O}$  denote the real octonion algebra with basis  $\{e_0, e_1, \dots, e_7\}$ , where  $e_0$  is the identity. The multiplication is determined by the seven *Fano triples*

$$(1) \quad (1, 2, 4), (2, 3, 5), (3, 4, 6), (4, 5, 7), (5, 6, 1), (6, 7, 2), (7, 1, 3),$$

encoding the rule: for each triple  $(a, b, c)$ ,  $e_a \cdot e_b = +e_c$ , with cyclic permutations carrying a positive sign and anti-cyclic permutations carrying a negative sign [1]. Every imaginary unit squares to  $-e_0$ . The algebra  $\mathbb{O}$  is the unique real normed division algebra of dimension 8 [1].

The *norm form* is  $N(x) = x\bar{x} = \sum_{i=0}^7 x_i^2$  for  $x = \sum_i x_i e_i$ , where  $\bar{x} = x_0 e_0 - \sum_{i=1}^7 x_i e_i$  is the conjugate. The composition property  $N(xy) = N(x)N(y)$  holds [1].

**2.2. The  $E_8$  lattice.** Wilson's  $E_8$  lattice is  $L = D_8^+$ , the set of vectors in  $\mathbb{R}^8$  whose coordinates (in the basis  $\{e_0, \dots, e_7\}$ ) are either all integers with even coordinate sum, or all half-integers with odd coordinate sum [22, 5]. It has 240 roots (vectors of squared norm 2). Under the standard octonion multiplication,  $L$  is a maximal order in  $\mathbb{O}$ : it is closed under multiplication [6, 5].

*Remark 2.1* (Kirmse's convention versus ours). The closure  $L \cdot L \subseteq L$  depends on which Fano-plane labelling defines the octonion product. Kirmse [20] originally proposed a specific labelling under which  $D_8^+$  is *not* closed — a subtle error corrected by Coxeter [6], who showed that permuting two basis indices yields a different multiplication on the same  $\mathbb{R}^8$  under which  $D_8^+$  *is* closed. There are seven such Coxeter-corrected conventions, each giving one of the seven maximal orders of the integral octonions [5]; see also Petersson [21] for a modern account.

The “standard” Fano triples (1) used in this paper are one of the Coxeter-corrected conventions, not Kirmse's. We have verified  $L \cdot L \subseteq L$  directly under our convention: using an explicit  $\mathbb{Z}$ -basis of  $L$  and exact rational arithmetic, all 64 basis products are integer combinations of the basis.

*Remark 2.2* (Closure of the lattice, not the shells). When we say  $L$  is closed under multiplication, we mean closure of the lattice  $L$ , not of any individual shell. Since the octonion product is norm-multiplicative, the product of two roots (squared norm 2) has squared norm 4 and lies on the second shell of  $L$ , not among the roots.

The same distinction will apply to  $(\Lambda, +, \star)$  in Theorem 1.1: products of minimal Leech vectors land on higher shells of  $\Lambda$ , not back in the minimal shell (see Section 5).

**2.3. Wilson's Leech lattice.** Define  $s = \frac{1}{2}(-e_0 + e_1 + \dots + e_7)$  and  $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \dots - e_7)$ , both of norm  $N(s) = N(\bar{s}) = 2$ . Then  $s \in L$  (it is a type-2 root, with an odd coordinate sum), while  $\bar{s} \notin L$  (its coordinate sum is  $-4$ , which falls in the excluded even branch of the half-integer case). Let

$$L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}, \quad Ls = \{\ell \cdot s : \ell \in L\}.$$

Although  $\bar{s}$  itself does not lie in  $L$ , the set  $L\bar{s}$  nonetheless lies in  $L$ : for every  $\ell \in L$  the product  $\ell \cdot \bar{s}$  lies in  $L$  (directly verified on the 240 roots of  $L$  and extended by linearity). Both  $L\bar{s}$  and  $Ls$  are therefore sublattices of  $L$ , with  $[L : L\bar{s}] = [L : Ls] = 16$ ,  $L\bar{s} + Ls = L$ , and  $L\bar{s} \cap Ls = 2L$  [22].

**Definition 2.3** (Wilson [22, Section 3]). The *Leech lattice* is

$$\Lambda = \{(x, y, z) \in L^3 : (1) x, y, z \in L; (2) x+y, x+z, y+z \in L\bar{s}; (3) x+y+z \in Ls\}.$$

The 196,560 minimal vectors of  $\Lambda$  decompose into three families [22]:

| Family | Formula                                                    | Count                                |
|--------|------------------------------------------------------------|--------------------------------------|
| Type 1 | $(2\lambda, 0, 0)$ and cyclic permutations                 | $3 \times 240 = 720$                 |
| Type 2 | $(\lambda\bar{s}, (\lambda\bar{s})j, 0)$ and cyclic perms  | $3 \times 240 \times 16 = 11,520$    |
| Type 3 | $((\lambda s)j, \lambda k, (\lambda j)k)$ and cyclic perms | $3 \times 240 \times 16^2 = 184,320$ |

Here  $\lambda$  ranges over the 240 roots of  $L$ , and  $j, k \in J = \{\pm e_t : 0 \leq t \leq 7\}$  (16 unit octonions).

### 3. THE CONSTRUCTION

**Definition 3.1** (Transposition-twisted octonion algebra). Let  $\sigma = (s \ t)$  be a transposition on  $\{1, \dots, 7\}$ . Extend  $\sigma$  to a linear map  $\sigma : \mathbb{R}^8 \rightarrow \mathbb{R}^8$  that fixes  $e_0$  and permutes  $e_s \leftrightarrow e_t$ . Apply  $\sigma$  to each standard Fano triple:  $(a, b, c) \mapsto (\sigma(a), \sigma(b), \sigma(c))$ . The resulting triples define a new multiplication table on  $\mathbb{R}^8$ . Equivalently, the  $\sigma$ -*twisted multiplication* is

$$(2) \quad x \cdot_\sigma y = \sigma(\sigma(x) \cdot \sigma(y)),$$

where  $\cdot$  denotes the standard octonion product.

**Proposition 3.2.**  $(\mathbb{R}^8, \cdot_\sigma)$  is an octonion algebra, isomorphic to  $(\mathbb{O}, \cdot)$  via the map  $\sigma$  itself:

$$(3) \quad \sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y) \quad \text{for all } x, y \in \mathbb{R}^8.$$

In particular,  $\cdot_\sigma$  satisfies the composition property  $N(x \cdot_\sigma y) = N(x)N(y)$ .

*Proof.* For any  $x, y \in \mathbb{R}^8$ :

$$\sigma(x) \cdot_\sigma \sigma(y) = \sigma(\sigma(\sigma(x)) \cdot \sigma(\sigma(y))) = \sigma(x \cdot y),$$

since  $\sigma^2 = \text{id}$ . Thus  $\sigma : (\mathbb{O}, \cdot) \rightarrow (\mathbb{R}^8, \cdot_\sigma)$  is a vector-space isomorphism that preserves multiplication. Composition follows.  $\square$

*Remark 3.3.* For the transposition  $\sigma = (1 \ 2)$ , exactly 30 of the 64 multiplication table entries change relative to the standard table: 6 entries undergo a sign change only, 16 undergo a target-index change only, and 8 undergo both.

*Remark 3.4.* The 21 transpositions of  $\{1, \dots, 7\}$  form a single orbit under the Fano-plane automorphism group  $\text{GL}(3, \mathbb{F}_2)$  (which has order 168 and acts on the 7 points of the Fano plane). Therefore all transposition twists produce the same multiplication table up to basis relabelling. The construction is essentially unique.

**Definition 3.5** (Triple product with  $\mathbb{Z}_3$  routing). Decompose  $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$  into three copies of  $\mathbb{R}^8$  (blocks indexed  $\alpha = 1, 2, 3$ ). For  $u = (x, y, z)$  and  $v = (x', y', z')$  with  $x, x' \in \mathbb{O}_1$ , etc., define the *triple product*

$$(4) \quad u \star v = (P_1, P_2, P_3),$$

where each block  $P_\alpha$  is the sum of three  $\sigma$ -twisted octonion products:

$$(5) \quad P_1 = x \cdot_\sigma x' + z \cdot_\sigma y' + y \cdot_\sigma z',$$

$$(6) \quad P_2 = y \cdot_\sigma y' + x \cdot_\sigma z' + z \cdot_\sigma x',$$

$$(7) \quad P_3 = z \cdot_\sigma z' + y \cdot_\sigma x' + x \cdot_\sigma y'.$$

The routing rule is: a product of blocks  $\alpha$  and  $\beta$  lands in block  $\gamma$ , where  $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$  for  $\alpha \neq \beta$ , and in block  $\alpha$  when  $\alpha = \beta$ . All nine block-pair products use the same multiplication  $\cdot_\sigma$ .

#### 4. PROOF OF CLOSURE

We prove Theorem 1.1 by showing that the twisted product preserves each of Wilson's three conditions (Definition 2.3). The proof rests on four lemmas about the interaction between the transposition  $\sigma$ , the maximal order  $L$ , and the sublattices  $Ls$  and  $L\bar{s}$ .

**4.1. Reduction to standard products.** Since each  $\sigma$ -twisted product has the form  $a \cdot_\sigma b = \sigma(\sigma(a) \cdot \sigma(b))$ , and  $\sigma$  is linear, each block  $P_\alpha$  is  $\sigma$  applied to a sum of standard octonion products of  $\sigma$ -images of the inputs.

Write  $X = \sigma(x)$ ,  $Y = \sigma(y)$ ,  $Z = \sigma(z)$ ,  $X' = \sigma(x')$ ,  $Y' = \sigma(y')$ ,  $Z' = \sigma(z')$ . Then:

$$(8) \quad P_1 = \sigma(X \cdot X' + Z \cdot Y' + Y \cdot Z'),$$

$$(9) \quad P_2 = \sigma(Y \cdot Y' + X \cdot Z' + Z \cdot X'),$$

$$(10) \quad P_3 = \sigma(Z \cdot Z' + Y \cdot X' + X \cdot Y').$$

Taking sums:

- **Block sum.**  $P_1 + P_2 + P_3 = \sigma((X + Y + Z) \cdot (X' + Y' + Z'))$ .
- **Pairwise sums.** For example,  $P_1 + P_2 = \sigma(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y'))$ , and similarly for the other two pairs.

#### 4.2. The four lemmas.

**Lemma 4.1.**  $\sigma(L) = L$ .

*Proof.* The lattice  $L = D_8^+$  consists of all vectors in  $\mathbb{R}^8$  whose coordinates are either all integers or all half-integers, with even coordinate sum. The transposition  $\sigma$  permutes two coordinates, which preserves both the integer/half-integer parity and the coordinate sum. Hence  $\sigma(L) \subseteq L$ . Since  $\sigma$  is an involution,  $\sigma(L) = L$ .  $\square$

For each of the next three lemmas,  $L$ ,  $\sigma(L\bar{s})$ , and  $\sigma(Ls)$  are full-rank  $\mathbb{Z}$ -lattices of rank 8 in  $\mathbb{R}^8$ . By  $\mathbb{Z}$ -bilinearity of the octonion product, an inclusion  $A \cdot B \subseteq C$  between such lattices holds if and only if it holds on a chosen pair of  $\mathbb{Z}$ -bases: if  $\{a_i\}_{i=1}^8$  generates  $A$  and  $\{b_j\}_{j=1}^8$  generates  $B$ , then  $A \cdot B \subseteq C$  is equivalent to each of the 64 products  $a_i \cdot b_j$  lying in  $\mathbb{Z}\langle c_1, \dots, c_8 \rangle$  for a chosen  $\mathbb{Z}$ -basis  $\{c_k\}_{k=1}^8$  of  $C$ . Each lemma below is proved by exhibiting that finite list of integer coefficients explicitly. Concrete  $\mathbb{Z}$ -bases for  $L$ ,  $\sigma(Ls)$ , and  $\sigma(L\bar{s})$  are fixed in Appendix A (denoted there  $L_k$ ,  $M_k$ ,  $N_k$  respectively); the 64-entry coefficient tables for the three lemmas appear as Tables 1, 2, and 3.

**Lemma 4.2.**  $L \cdot L \subseteq L$ .

*Proof.* The 64 products  $L_i \cdot L_j$  are integer combinations of  $\{L_1, \dots, L_8\}$ ; the explicit coefficients are Table 1. By  $\mathbb{Z}$ -bilinearity,  $L \cdot L \subseteq L$ . This recovers the classical fact that  $L = D_8^+$  is a maximal order in  $\mathbb{O}$  [6, 5].  $\square$

**Lemma 4.3.**  $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$ .

*Proof.* The 64 products  $L_i \cdot N_j$  are integer combinations of  $\{N_1, \dots, N_8\}$ ; the explicit coefficients are Table 2.  $\square$

**Lemma 4.4.**  $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ .

*Proof.* The 64 products  $M_i \cdot M_j$  are integer combinations of  $\{M_1, \dots, M_8\}$ ; the explicit coefficients are Table 3.  $\square$

*Remark 4.5.* The sublattice  $Ls$  is *not* closed under the standard octonion product: there exist basis vectors  $a, b \in Ls$  with  $a \cdot b \notin Ls$ . This is precisely the obstruction that the twist resolves: it moves the block-sum sublattice from  $Ls$  to  $\sigma(Ls)$ , where closure holds by Lemma 4.4.

Note that  $\sigma(Ls)$  is genuinely different from  $Ls$  (not just a relabelling). For example, with  $\sigma = (1\ 2)$ , the element  $v = (-1, 0, 1, 0, 0, 1, 0, 1)$  lies in  $\sigma(Ls)$  but direct computation shows it is not an integer combination of any  $\mathbb{Z}$ -basis for  $Ls$ , so  $v \notin Ls$ . This distinction is what makes the twist useful:  $\sigma$  is not the identity on the block-sum sublattice, so it can redirect a product into a different (and, as it turns out, closed) sublattice.

*Remark 4.6* (Two equivalent framings). Let  $\star_0$  denote the analogous triple product built from the *standard* (untwisted) octonion multiplication, i.e. with  $\cdot$  replacing  $\cdot_\sigma$  in every block-pair product of Definition 3.5. Applying  $\sigma$  block-wise gives an algebra isomorphism  $\Sigma := \sigma \oplus \sigma \oplus \sigma : (\mathbb{R}^{24}, \star) \rightarrow (\mathbb{R}^{24}, \star_0)$ , since each block  $P_\alpha$  of  $u \star v$  is by construction (8)–(10) the image under  $\sigma$  of the corresponding block of  $\Sigma(u) \star_0 \Sigma(v)$ . Consequently the central Theorem 1.1 has two equivalent readings:

$$\Lambda \star \Lambda \subseteq \Lambda \quad \Longleftrightarrow \quad \Sigma(\Lambda) \star_0 \Sigma(\Lambda) \subseteq \Sigma(\Lambda).$$

Neither reading asserts  $\Lambda \star_0 \Lambda \subseteq \Lambda$ , which fails: the standard triple product obstructs Wilson's condition (3), since it would require  $Ls \cdot Ls \subseteq Ls$ . The twist does not rename a closure that was already present; it identifies a distinct Leech embedding  $\Sigma(\Lambda) \neq \Lambda$  — distinct because  $\sigma(Ls) \neq Ls$  — on which the standard triple product closes.

#### 4.3. Condition (1): each block lies in $L$ .

**Proposition 4.7.** *For all  $u, v \in \Lambda$ , each block  $P_\alpha$  of  $u \star v$  lies in  $L$ .*

*Proof.* Consider  $P_1 = \sigma(X \cdot X' + Z \cdot Y' + Y \cdot Z')$ . Since  $u \in \Lambda$ , condition (1) gives  $x \in L$ , so  $X = \sigma(x) \in L$  by Lemma 4.1. Similarly  $X', Y, Y', Z, Z' \in L$ . By Lemma 4.2, each product  $X \cdot X', Z \cdot Y', Y \cdot Z'$  lies in  $L$ . Their sum lies in  $L$  (closed under addition). Applying  $\sigma$  and using Lemma 4.1,  $P_1 \in L$ . The same argument applies to  $P_2$  and  $P_3$ .  $\square$

#### 4.4. Condition (2): pairwise sums lie in $L\bar{s}$ .

**Proposition 4.8.** *For all  $u, v \in \Lambda$ , the pairwise sums  $P_\alpha + P_\beta$  lie in  $L\bar{s}$ .*

*Proof.* We show  $P_1 + P_2 \in L\bar{s}$ ; the other two cases follow by the  $\mathbb{Z}_3$  symmetry of the routing.

From (8)–(9):

$$P_1 + P_2 = \sigma(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')).$$

Since  $v \in \Lambda$ , condition (2) gives  $x' + z' \in L\bar{s}$ , so  $X' + Z' = \sigma(x' + z') \in \sigma(L\bar{s})$ . Similarly  $Y' + Z' \in \sigma(L\bar{s})$  and  $X' + Y' \in \sigma(L\bar{s})$ . Since  $u \in \Lambda$ , we have  $X, Y, Z \in L$  (by Lemma 4.1).

By Lemma 4.3, each product  $X \cdot (X' + Z'), Y \cdot (Y' + Z'), Z \cdot (X' + Y')$  lies in  $\sigma(L\bar{s})$ . Their sum lies in  $\sigma(L\bar{s})$ . Applying the outer  $\sigma$ :

$$P_1 + P_2 = \sigma(\text{element of } \sigma(L\bar{s})) \in \sigma(\sigma(L\bar{s})) = L\bar{s},$$

since  $\sigma^2 = \text{id}$ .  $\square$

**4.5. Condition (3): the block sum lies in  $Ls$ .** This is the crux of the proof. For the standard (untwisted) product the block sum would require  $Ls \cdot Ls \subseteq Ls$ , which does not hold (Remark 4.5). The twist redirects the computation through  $\sigma(Ls)$ , where closure holds.

**Proposition 4.9.** *For all  $u, v \in \Lambda$ , the block sum  $P_1 + P_2 + P_3$  lies in  $Ls$ .*

*Proof.* From (8)–(10):

$$P_1 + P_2 + P_3 = \sigma((X + Y + Z) \cdot (X' + Y' + Z')).$$

Since  $u \in \Lambda$ , condition (3) gives  $x + y + z \in Ls$ , so

$$X + Y + Z = \sigma(x + y + z) \in \sigma(Ls).$$

Similarly  $X' + Y' + Z' \in \sigma(Ls)$ .

By Lemma 4.4,  $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ , so

$$(X + Y + Z) \cdot (X' + Y' + Z') \in \sigma(Ls).$$

Applying the outer  $\sigma$ :

$$P_1 + P_2 + P_3 \in \sigma(\sigma(Ls)) = Ls. \quad \square$$

This completes the proof of Theorem 1.1.

## 5. ALGEBRAIC PROPERTIES

The order  $(\Lambda, +, \star)$  has been tested for several standard algebraic properties. The results are summarised below.

| Property                | Holds? | Details                                                                      |
|-------------------------|--------|------------------------------------------------------------------------------|
| Multiplicative identity | No     | the ambient algebra $(\mathbb{R}^{24}, +, \star)$ has no identity element    |
| Commutativity           | No     | $< 0.1\%$ of pairs commute                                                   |
| Norm multiplicativity   | No     | $\approx 47\%$ of pairs satisfy $N(u \star v) = N(u)N(v)$                    |
| Left alternativity      | No     | $< 0.1\%$ of pairs satisfy $(x \star x) \star y = x \star (x \star y)$       |
| Right alternativity     | No     | $< 0.1\%$ of pairs satisfy $(x \star y) \star y = x \star (y \star y)$       |
| Flexibility             | No     | $\approx 8.6\%$ of pairs satisfy $(x \star y) \star x = x \star (y \star x)$ |
| Power-associativity     | No     | $\approx 29\%$ of elements satisfy $x^2 \star x = x \star x^2$               |

*Remark 5.1.* The product norm on  $\text{Min}(\Lambda) \times \text{Min}(\Lambda)$  takes values in  $\{16, 32, 48, 64, 80, 96, 112, 128\}$ , all multiples of 16. The mode is 64 (the norm-multiplicative value  $N(u)N(v) = 8 \times 8$ ).

*Remark 5.2.* The ambient algebra  $(\mathbb{R}^{24}, +, \star)$  has no multiplicative identity: any putative left identity  $(a, b, c)$  would have to act on an arbitrary  $(x', y', z')$  by the block matrix  $\begin{pmatrix} a & c & b \\ c & b & a \\ b & a & c \end{pmatrix}$  (entries acting by left-multiplication in  $\mathbb{O}$ ), and no triple  $(a, b, c)$  makes this the identity on all three blocks simultaneously. In particular,  $(e_0, e_0, e_0)$  is not an identity: it maps  $(x', y', z')$  to  $(x' + y' + z', x' + y' + z', x' + y' + z')$ . Consequently  $(\Lambda, +, \star)$  is a non-unital order.

The order does not satisfy any of the classical algebraic identities (associativity, alternativity, flexibility, power-associativity, commutativity). The defining property is *closure*: the Leech lattice is stable under the product. The relationship between  $\star$  and the Conway group  $\text{Co}_0$  (the automorphism group of  $\Lambda$  as a lattice) is an open question.

## 6. RESEARCH METHODOLOGY

This research was conducted as a collaboration between a human researcher and an AI agent (Claude, Anthropic) over a series of directed prompts, governed by a written methodology (the “Manifesto”) established at the outset.



The human researcher (the author) brought the initial hypothesis — motivated by prior work on Okubo algebras and autotopies of normed composition algebras — established the research methodology, and maintained control of the inquiry through prompt-by-prompt review. The AI agent performed the computational work: implementing octonion algebras, Wilson’s Leech lattice construction, trial algebras, and closure tests, and contributed mathematical analysis.

Tests were written from first principles to independently verify every foundation: that octonion multiplication satisfies the composition property  $N(xy) = N(x)N(y)$ , that Wilson’s construction produces a lattice with the correct minimal shell of 196,560 vectors, and that the  $E_8$  lattice is indeed a maximal order. This independent verification is how we know that the properties stated in the literature are correctly implemented. Closure of the twisted product was first observed on over 12,000,000 random pairs of minimal vectors with zero failures, which prompted the search for a symbolic proof.

Symbolic proof and numeric validation. The proof of Theorem 1.1 given in Section 4 is symbolic: each of the four lemmas is established by exhibiting a finite list of integer coefficients (Appendix A), and the conclusion follows by  $\mathbb{Z}$ -bilinearity. No floating-point or sampled arithmetic enters the proof. Independently, the same basis-by-basis identities that constitute the appendix tables were recomputed in two separate computer-algebra implementations as a cross-check: a Python implementation using `fractions.Fraction` exact rationals (`python.project/src/symbolic_proof`) and a GAP/LOOPS implementation using GAP’s native exact rational arithmetic (`gap_project/tests/test_lemmas.g`). Both implementations reproduce the same integer coefficients displayed in the appendix. This redundancy is validation, not proof: it serves to detect transcription or implementation errors in the appendix tables, but the logical content of the proof rests on the tables themselves.

The methodology required:

- Every claim backed by a reference or computation.
- Every prompt logged, with actions recorded.
- Every conclusion independently verified.
- Forward-evolving corrections: no rewriting history; errors are corrected in subsequent entries.

The path to the result was not direct. Six trial algebra families were explored and ruled out before the transposition twist was identified in trial 007. The systematic methodology — understanding the anatomy of failure (which conditions fail, for which vector types, and why) before trying the next approach — was essential. The failure analysis of the untwisted product (Wilson’s condition (3) per Definition 2.3, exclusively type-3×type-3) directly pointed to the mechanism that the twist needed to correct.

Iterative convergence toward a robust result — including learning along the way what it is we are actually doing — is how study, research, and discovery have always proceeded. The AI agent violated its own logging protocol on three occasions (prompts 011, 018–019, and 029–031), each time caught by the human researcher on review. Conversely, the author learned from the AI agent about errors and misconceptions in the framing of certain prompts, which likewise are part of the record. What is novel here is the ability to trace the entire path, beginning to end, through the complete list of prompts.

The full record — all prompts, code, evidence, and git history — is available at the repository [23]. The reader can form their own assessment of what this means for questions of discovery and attribution.

## 7. RELATED WORK

**Wilson (2009)** [22] characterises  $\Lambda$  as  $\{(x, y, z) \in L^3 : \text{conditions 1–3}\}$ , gives an explicit listing of the 196,560 minimal vectors in the form  $3 \times 240 \times (1 + 16 + 16 \times 16)$ , and proves that  $\Lambda$  is even and self-dual. The sublattice framework is the setting used in the present paper.

**Dixon (2010)** [8] constructs  $\Lambda$  using the octonion XY-product<sup>1</sup>

$$A \circ_{XY} B = (AX)(Y^\dagger B),$$

where  $X, Y \in \mathbb{O}$  are chosen so that  $YX^\dagger$  is a unit octonion (the identity of the new multiplication). When  $Y^\dagger = X^{-1}$  the XY-product specialises to the X-product  $A \circ_X B = (AX)(X^{-1}B)$ ; Dixon uses this X-product with  $X = \ell_0 = \frac{1}{2}(1 + e_1 + \cdots + e_7)$  to obtain  $E_8$ , and general XY-products for higher-dimensional laminated lattices including  $\Lambda$ .

**Baez and Egan (2014)** [2] study closure properties of integer sublattices of the exceptional Jordan algebra  $\mathfrak{h}_3(\mathbb{O})$ . In parts 9–10 of the series, Baez considers the 27-dimensional lattice  $\mathbb{Z}^3 \oplus L_L \subset \mathfrak{h}_3(\mathbb{O})$ , with integers on the diagonal and an off-diagonal Leech sublattice  $L_L \cong \Lambda$  embedded as an octonionic triple. This lattice is *not* closed under the Jordan product  $x \circ y = xy + yx$ , but *is* closed under the doubled product  $2(xy + yx)$ . Egan further enumerated at least 17,280 distinct off-diagonal Leech embeddings  $L_L \subset \mathbb{O}^3$  for which the associated 27-dimensional lattice is closed under this doubled product. Baez left open the question whether any such lattice is closed under the undoubled Jordan product — which would remove the factor-of-2 discrepancy — and conjectures that the answer is negative.

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<sup>1</sup>The construction has an earlier precursor in Dixon [7], where a closely related octonionic description of  $\Lambda$  is proposed. The formula there required a small correction in the third family of minimal vectors, which Wilson [22, end of §5] supplied in a footnote-style remark while independently deriving the sublattice characterisation. The situation is similar to the Kirmse–Coxeter relationship discussed below: an early, structurally correct proposal needing correction.

**Earlier precursors.** Further back, several constructions foreshadow the octonion-triple and Jordan-algebra frameworks above. Conway and Sloane [3] give 23 constructions of  $\Lambda$ , one for each Niemeier lattice, including a Turyn-style construction from three copies of  $E_8$ . Lepowsky and Meurman [17] describe  $\Lambda$  as a sublattice of  $E_8^3$  containing  $2E_8^3$  and use this to generate the Conway group  $\text{Co}_0$  via permutations of the three  $E_8$  factors together with one extra automorphism. Elkies and Gross [12] obtain  $\Lambda$  from an integer form of the exceptional Jordan algebra  $\mathfrak{h}_3(\mathbb{O})$  as the orthogonal complement of a distinguished identity element, and show that the automorphism group of this integer form is generated by octave reflections. Each of these constructions is additive or geometric: none defines or tests a bilinear product on  $\Lambda$ .

**Kirmse (1924)** [20] exhibited the  $E_8$  lattice as a unimodular integral quadratic lattice inside the real octonion algebra. His lattice is not closed under multiplication, an error corrected by Coxeter [6] by swapping two basis indices in the Fano-plane labelling.

The present construction continues the same conceptual line. The Kirmse–Coxeter step modifies the octonion multiplication (a single transposition of basis indices) so that the  $E_8$  lattice becomes an order. The present paper applies a second transposition of basis indices — the twist  $\sigma$  in Definition 3.1 — to the already-corrected octonion product, so that the triple-product construction on the Leech lattice becomes an order. The chain is Kirmse  $\rightarrow$  Coxeter  $\rightarrow$   $\sigma$ -twist: each step is a transposition, and each resolves a closure obstruction at the next level of the construction.

## 8. CONCLUSION

The Leech lattice admits an order under a remarkably simple construction: a  $\mathbb{Z}_3$ -symmetric triple-octonion product on  $\mathbb{R}^{24}$ , realised in the Coxeter representation aligned with Wilson’s sublattice description of  $\Lambda$ . The symbolic proof (Section 4) reduces closure to four lemmas, of which the crux is Lemma 4.4: the sublattice  $\sigma(Ls)$  — unlike  $Ls$  itself — is closed under the standard octonion product. The twist maps the condition-3 sublattice to this closed sublattice, and the involutory nature of  $\sigma$  maps the result back.

Several questions remain immediately open:

- A structural reason for Lemma 4.4. The proof in Appendix A exhibits the 64 integer coefficients certifying  $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ , but does not explain *why*  $\sigma(Ls)$  is closed under the standard octonion product while  $Ls$  is not. An identification of  $\sigma(Ls)$  with a left ideal, with the image under a norm-preserving octonion automorphism, or with some module-theoretic structure over  $L$ , would replace the appendix tables by a structural argument.

- The automorphism group of  $(\Lambda, +, \star)$  and its relationship to the Conway group  $\text{Co}_0$ .
- Whether the order is maximal in any appropriate sense.
- Classification of the algebra  $(\mathbb{R}^{24}, +, \star)$ : it is non-associative, non-alternative, non-flexible, and non-unital (as an order), placing it outside the classical families.
- Whether other permutations of the Fano triples (beyond transpositions) also yield closed products on  $\Lambda$ .

## 9. OUTLOOK

The intention in publishing this result at this stage is to invite research rather than to sit on the ideas. If others arrive at deeper answers faster, so much the better. What follows is an outlook that names the directions we find most promising and the sources that shaped the intuition leading to the present construction.

**9.1. A ternary reformulation.** A more natural classification of  $(\mathbb{R}^{24}, +, \star)$  may emerge from a *ternary* rather than binary viewpoint. The Leech lattice possesses a prominent  $\mathbb{Z}_3$ -symmetry (the cyclic permutation of the three octonion blocks used throughout this paper), and the binary product  $\star$  is assembled from nine bilinear building blocks whose routing is itself governed by that  $\mathbb{Z}_3$ -action. Ternary composition algebras, as studied by Elduque [9, 10, 11], are typically non-associative, non-commutative, non-alternative, and non-power-associative — the same negative classification our order satisfies — yet carry high internal symmetry, with order-3 automorphisms playing the central role: the Okubo algebra has automorphism group  $\text{SU}(3)$ , and the Okubo product itself is built from an order-3 automorphism of the octonions. Complementary perspectives on the Okubo algebra and triality include the quasigroup-theoretic treatment of Smith and Vojtěchovský [19], Kamiya and Okubo’s study of algebras satisfying triality and  $S_4$ -symmetry [15], and Okubo’s seminal book [18]. A deeper precursor is Hall’s monograph [14], which establishes that Moufang loops and groups with triality are essentially the same object, giving a structural scaffolding for any triality-based reformulation. Recasting  $\star$  as a trace-like reduction of a ternary operation compatible with the  $\mathbb{Z}_3$ -block symmetry is a natural direction for further work.

**9.2. Consequences for algebraic approaches to physics.** A classification of the ambient algebra  $(\mathbb{R}^{24}, +, \star)$  matters beyond pure algebra. Programmes that seek to model fundamental physics on algebraic structures — especially those that use the four normed division algebras, triality, and related symmetric composition algebras — have grown considerably in recent years. The author previously invited [16] for research into autotopies of normed composition algebras as a mathematical setting for this line of inquiry. Recent work of Furey and

Hughes [13] isolates triality itself as the operative structural skeleton: the Standard Model’s  $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$  gauge symmetries are embedded inside  $\mathfrak{tri}(\mathbb{C}) \oplus \mathfrak{tri}(\mathbb{H}) \oplus \mathfrak{tri}(\mathbb{O})$ , and the triality skeleton is shown to accommodate a variety of particle-family representations — the authors’ own assignment of three Standard Model generations to a triality triple in  $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$  is one concrete realization of that wider scaffolding.

**9.3. On the form of this work.** This paper is the outcome of a human–AI collaboration conducted entirely through a sequence of AI prompts (Section 6), with the complete record of prompts, code, intermediate results, and corrections preserved verbatim [23]. The form itself is part of the contribution. Collaborations of this kind will become more common, and this work is offered in its present form partly in the hope that the process — how a research question was framed, how the AI was directed and corrected, how false trails were recognised and closed — is instructive for mathematicians considering AI-assisted research. Comments on both substance and method are welcome.

#### ACKNOWLEDGMENTS

The author thanks Geoffrey M. Dixon and Robert A. Wilson for helpful discussions and advice on octonionic structures associated with the Leech lattice.

#### APPENDIX A. EXPLICIT BASIS TABLES FOR LEMMAS 4.2–4.4

We fix concrete  $\mathbb{Z}$ -bases for the lattices  $L$ ,  $\sigma(Ls)$ , and  $\sigma(L\bar{s})$ , and tabulate the integer coefficients of the basis-by-basis products called for in the proofs of Lemmas 4.2, 4.3, and 4.4. Each table has 64 entries and is arranged so that the row labelled  $(i, j)$  gives the unique 8-tuple  $(c_1, \dots, c_8) \in \mathbb{Z}^8$  representing the corresponding product in the basis indicated. All entries are integer; closure under the standard octonion product follows by  $\mathbb{Z}$ -bilinearity. The same identities have been re-computed in two independent computer-algebra implementations as a cross-check (Section 6).

Coordinates throughout are with respect to the basis  $\{e_0, \dots, e_7\}$  and the standard Fano triples (1).

The basis  $\{L_k\}$ . The first vector is Wilson’s  $s = \frac{1}{2}(-e_0 + e_1 + \dots + e_7)$ ; the remaining seven are type-1 roots of  $L$  chosen so that  $\{L_1, \dots, L_8\}$

is linearly independent over  $\mathbb{Q}$  and spans  $L$  over  $\mathbb{Z}$ .

$$\begin{aligned}
L_1 &= \left(\frac{-1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \\
L_2 &= (1, 1, 0, 0, 0, 0, 0, 0) \\
L_3 &= (1, -1, 0, 0, 0, 0, 0, 0) \\
L_4 &= (1, 0, 1, 0, 0, 0, 0, 0) \\
L_5 &= (1, 0, 0, 1, 0, 0, 0, 0) \\
L_6 &= (1, 0, 0, 0, 1, 0, 0, 0) \\
L_7 &= (1, 0, 0, 0, 0, 1, 0, 0) \\
L_8 &= (1, 0, 0, 0, 0, 0, 1, 0)
\end{aligned}$$

The basis  $\{M_k\}$ . Defined by  $M_k := \sigma(L_k \cdot s)$  with  $\sigma = (1\ 2)$  acting on coordinates by swapping  $e_1$  and  $e_2$ . Spans  $\sigma(Ls)$  over  $\mathbb{Z}$ .

$$\begin{aligned}
M_1 &= \left(\frac{-3}{2}, \frac{-1}{2}, \frac{-1}{2}, \frac{-1}{2}, \frac{-1}{2}, \frac{-1}{2}, \frac{-1}{2}, \frac{-1}{2}\right) \\
M_2 &= (-1, 0, 0, 0, 1, 0, 1, 1) \\
M_3 &= (0, 1, 1, 1, 0, 1, 0, 0) \\
M_4 &= (-1, 0, 1, 0, 0, 1, 0, 1) \\
M_5 &= (-1, 1, 1, 0, 0, 0, 1, 0) \\
M_6 &= (-1, 1, 0, 1, 0, 0, 0, 1) \\
M_7 &= (-1, 0, 1, 1, 1, 0, 0, 0) \\
M_8 &= (-1, 1, 0, 0, 1, 1, 0, 0)
\end{aligned}$$

The basis  $\{N_k\}$ . Defined by  $N_k := \sigma(L_k \cdot \bar{s})$ . Spans  $\sigma(L\bar{s})$  over  $\mathbb{Z}$ .

$$\begin{aligned}
N_1 &= (2, 0, 0, 0, 0, 0, 0, 0) \\
N_2 &= (0, 0, -1, 0, -1, 0, -1, -1) \\
N_3 &= (-1, -1, 0, -1, 0, -1, 0, 0) \\
N_4 &= (0, -1, -1, 0, 0, -1, 0, -1) \\
N_5 &= (0, -1, -1, -1, 0, 0, -1, 0) \\
N_6 &= (0, -1, 0, -1, -1, 0, 0, -1) \\
N_7 &= (0, 0, -1, -1, -1, -1, 0, 0) \\
N_8 &= (0, -1, 0, 0, -1, -1, -1, 0)
\end{aligned}$$

**Lemma 4.2:**  $L \cdot L \subseteq L$ .

| $i$ | $j$ | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ |
|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|
| 1   | 1   | -1    | -1    | -1    | 0     | 0     | 0     | 0     | 0     |
| 1   | 2   | 0     | -2    | -2    | 1     | 1     | 0     | 1     | 0     |
| 1   | 3   | 2     | 2     | 2     | -1    | -1    | 0     | -1    | 0     |
| 1   | 4   | 0     | -2    | -2    | 0     | 1     | 1     | 0     | 1     |
| 1   | 5   | 2     | 1     | 2     | -1    | -1    | 0     | 0     | -1    |
| 1   | 6   | 0     | -1    | -2    | 0     | 0     | 0     | 1     | 1     |
| 1   | 7   | 2     | 1     | 2     | 0     | -1    | -1    | -1    | 0     |
| 1   | 8   | 2     | 2     | 2     | -1    | 0     | -1    | -1    | -1    |
| 2   | 1   | 2     | 1     | 2     | -1    | -1    | 0     | -1    | 0     |
| 2   | 2   | 0     | 1     | -1    | 0     | 0     | 0     | 0     | 0     |
| 2   | 3   | 0     | 1     | 1     | 0     | 0     | 0     | 0     | 0     |
| 2   | 4   | 0     | 0     | -1    | 1     | 0     | 1     | 0     | 0     |
| 2   | 5   | 2     | 3     | 3     | -1    | 0     | -1    | -1    | -1    |
| 2   | 6   | 0     | 1     | 0     | -1    | 0     | 1     | 0     | 0     |
| 2   | 7   | 0     | 0     | -1    | 0     | 0     | 0     | 1     | 1     |
| 2   | 8   | 0     | 1     | 0     | 0     | 0     | 0     | -1    | 1     |
| 3   | 1   | 0     | -1    | -2    | 1     | 1     | 0     | 1     | 0     |
| 3   | 2   | 0     | 1     | 1     | 0     | 0     | 0     | 0     | 0     |
| 3   | 3   | 0     | -1    | 1     | 0     | 0     | 0     | 0     | 0     |
| 3   | 4   | 0     | 0     | 1     | 1     | 0     | -1    | 0     | 0     |
| 3   | 5   | -2    | -3    | -3    | 1     | 2     | 1     | 1     | 1     |
| 3   | 6   | 0     | -1    | 0     | 1     | 0     | 1     | 0     | 0     |
| 3   | 7   | 0     | 0     | 1     | 0     | 0     | 0     | 1     | -1    |
| 3   | 8   | 0     | -1    | 0     | 0     | 0     | 0     | 1     | 1     |
| 4   | 1   | 2     | 2     | 2     | -1    | -1    | -1    | 0     | -1    |
| 4   | 2   | 0     | 1     | 0     | 1     | 0     | -1    | 0     | 0     |
| 4   | 3   | 0     | -1    | 0     | 1     | 0     | 1     | 0     | 0     |
| 4   | 4   | 0     | -1    | -1    | 2     | 0     | 0     | 0     | 0     |
| 4   | 5   | 0     | -1    | -1    | 1     | 1     | 0     | 1     | 0     |
| 4   | 6   | 0     | 0     | -1    | 1     | 0     | 1     | 0     | 0     |
| 4   | 7   | 0     | 0     | 0     | 1     | -1    | 0     | 1     | 0     |
| 4   | 8   | 2     | 2     | 3     | 0     | -1    | -1    | -1    | 0     |
| 5   | 1   | 0     | -1    | -2    | 1     | 0     | 0     | 0     | 1     |
| 5   | 2   | -2    | -2    | -4    | 1     | 2     | 1     | 1     | 1     |
| 5   | 3   | 2     | 2     | 4     | -1    | 0     | -1    | -1    | -1    |
| 5   | 4   | 0     | 0     | 0     | 1     | 1     | 0     | -1    | 0     |
| 5   | 5   | 0     | -1    | -1    | 0     | 2     | 0     | 0     | 0     |
| 5   | 6   | 0     | -1    | -1    | 0     | 1     | 1     | 0     | 1     |
| 5   | 7   | 0     | -1    | -1    | 1     | 1     | 0     | 1     | 0     |
| 5   | 8   | 0     | 0     | 0     | 0     | 1     | -1    | 0     | 1     |
| 6   | 1   | 2     | 1     | 2     | 0     | 0     | -1    | -1    | -1    |
| 6   | 2   | 0     | 0     | -1    | 1     | 0     | 1     | 0     | 0     |
| 6   | 3   | 0     | 0     | 1     | -1    | 0     | 1     | 0     | 0     |
| 6   | 4   | 0     | -1    | 0     | 1     | 0     | 1     | 0     | 0     |
| 6   | 5   | 0     | 0     | 0     | 0     | 1     | 1     | 0     | -1    |
| 6   | 6   | 0     | -1    | -1    | 0     | 0     | 2     | 0     | 0     |
| 6   | 7   | 2     | 2     | 3     | -1    | -1    | 0     | 0     | -1    |
| 6   | 8   | 0     | -1    | -1    | 0     | 1     | 1     | 0     | 1     |
| 7   | 1   | 0     | -1    | -2    | 0     | 1     | 1     | 0     | 0     |
| 7   | 2   | 0     | 1     | 0     | 0     | 0     | 0     | 1     | -1    |
| 7   | 3   | 0     | -1    | 0     | 0     | 0     | 0     | 1     | 1     |
| 7   | 4   | 0     | -1    | -1    | 1     | 1     | 0     | 1     | 0     |
| 7   | 5   | 0     | 0     | 0     | -1    | 1     | 0     | 1     | 0     |
| 7   | 6   | -2    | -3    | -4    | 1     | 1     | 2     | 2     | 1     |
| 7   | 7   | 0     | -1    | -1    | 0     | 0     | 0     | 2     | 0     |
| 7   | 8   | 0     | 0     | -1    | 0     | 0     | 0     | 1     | 1     |
| 8   | 1   | 0     | -2    | -2    | 1     | 0     | 1     | 1     | 0     |
| 8   | 2   | 0     | 0     | -1    | 0     | 0     | 0     | 1     | 1     |
| 8   | 3   | 0     | 0     | 1     | 0     | 0     | 0     | -1    | 1     |
| 8   | 4   | -2    | -3    | -4    | 2     | 1     | 1     | 1     | 2     |
| 8   | 5   | 0     | -1    | -1    | 0     | 1     | 1     | 0     | 1     |
| 8   | 6   | 0     | 0     | 0     | 0     | -1    | 1     | 0     | 1     |
| 8   | 7   | 0     | -1    | 0     | 0     | 0     | 0     | 1     | 1     |
| 8   | 8   | 0     | -1    | -1    | 0     | 0     | 0     | 0     | 2     |

TABLE 1. Coefficients  $c_k$  such that  $L_i \cdot L_j = \sum_{k=1}^8 c_k L_k$ , for the bases defined above.

**Lemma 4.3:**  $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$ .

| $i$ | $j$ | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ |
|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|
| 1   | 1   | -1    | -1    | -1    | 0     | 0     | 0     | 0     | 0     |
| 1   | 2   | 0     | -2    | -2    | 0     | 1     | 1     | 0     | 1     |
| 1   | 3   | 2     | 2     | 2     | 0     | -1    | -1    | 0     | -1    |
| 1   | 4   | 0     | -1    | -2    | 0     | 1     | 0     | 1     | 0     |
| 1   | 5   | 2     | 1     | 2     | -1    | -1    | 0     | 0     | -1    |
| 1   | 6   | 0     | -2    | -2    | 1     | 0     | 0     | 1     | 1     |
| 1   | 7   | 2     | 2     | 2     | -1    | -1    | -1    | -1    | 0     |
| 1   | 8   | 2     | 1     | 2     | 0     | 0     | -1    | -1    | -1    |
| 2   | 1   | 2     | 2     | 2     | -1    | -1    | -1    | 0     | -1    |
| 2   | 2   | -1    | 0     | -2    | 0     | 0     | 1     | 0     | 1     |
| 2   | 3   | 1     | 1     | 2     | 0     | 0     | 0     | -1    | 0     |
| 2   | 4   | 0     | 0     | -1    | 1     | 0     | 0     | 0     | 1     |
| 2   | 5   | 0     | 0     | -1    | 0     | 1     | 1     | 0     | 0     |
| 2   | 6   | 1     | 1     | 1     | 0     | -1    | 1     | -1    | 0     |
| 2   | 7   | 1     | 2     | 2     | -1    | -1    | 0     | 0     | 0     |
| 2   | 8   | 1     | 1     | 1     | -1    | 0     | 0     | -1    | 1     |
| 3   | 1   | 0     | -2    | -2    | 1     | 1     | 1     | 0     | 1     |
| 3   | 2   | 1     | 2     | 2     | 0     | 0     | -1    | 0     | -1    |
| 3   | 3   | -1    | -1    | 0     | 0     | 0     | 0     | 1     | 0     |
| 3   | 4   | 0     | 0     | 1     | 1     | 0     | 0     | 0     | -1    |
| 3   | 5   | 0     | 0     | 1     | 0     | 1     | -1    | 0     | 0     |
| 3   | 6   | -1    | -1    | -1    | 0     | 1     | 1     | 1     | 0     |
| 3   | 7   | -1    | -2    | -2    | 1     | 1     | 0     | 2     | 0     |
| 3   | 8   | -1    | -1    | -1    | 1     | 0     | 0     | 1     | 1     |
| 4   | 1   | 2     | 1     | 2     | -1    | -1    | 0     | -1    | 0     |
| 4   | 2   | 0     | 0     | -1    | 1     | 0     | 1     | 0     | 0     |
| 4   | 3   | 0     | 0     | 1     | 1     | 0     | -1    | 0     | 0     |
| 4   | 4   | 0     | -1    | -1    | 2     | 0     | 0     | 0     | 0     |
| 4   | 5   | 2     | 2     | 3     | 0     | 0     | -1    | -1    | -1    |
| 4   | 6   | 0     | -1    | 0     | 1     | 0     | 1     | 0     | 0     |
| 4   | 7   | 0     | -1    | -1    | 1     | 0     | 0     | 1     | 1     |
| 4   | 8   | 0     | 0     | 0     | 1     | 0     | 0     | -1    | 1     |
| 5   | 1   | 0     | -1    | -2    | 1     | 0     | 0     | 0     | 1     |
| 5   | 2   | 0     | 1     | 0     | 0     | 1     | 0     | -1    | 0     |
| 5   | 3   | 0     | -1    | 0     | 0     | 1     | 0     | 1     | 0     |
| 5   | 4   | -2    | -3    | -4    | 2     | 2     | 1     | 1     | 1     |
| 5   | 5   | 0     | -1    | -1    | 0     | 2     | 0     | 0     | 0     |
| 5   | 6   | 0     | -1    | -1    | 0     | 1     | 1     | 0     | 1     |
| 5   | 7   | 0     | 0     | -1    | 0     | 1     | 0     | 1     | 0     |
| 5   | 8   | 0     | 0     | 0     | 0     | 1     | -1    | 0     | 1     |
| 6   | 1   | 2     | 1     | 2     | 0     | 0     | -1    | -1    | -1    |
| 6   | 2   | 1     | 2     | 1     | -1    | 0     | 0     | 0     | -1    |
| 6   | 3   | -1    | -1    | -1    | 1     | 0     | 1     | 1     | 0     |
| 6   | 4   | 0     | 1     | 0     | 1     | 0     | 0     | 0     | -1    |
| 6   | 5   | 0     | 0     | 0     | 0     | 1     | 0     | 1     | -1    |
| 6   | 6   | -1    | -2    | -3    | 1     | 1     | 2     | 1     | 0     |
| 6   | 7   | 1     | 1     | 1     | 0     | -1    | 0     | 1     | -1    |
| 6   | 8   | 1     | 1     | 1     | 0     | 0     | 0     | 0     | 0     |
| 7   | 1   | 0     | -1    | -2    | 0     | 1     | 1     | 0     | 0     |
| 7   | 2   | -1    | -1    | -2    | 0     | 1     | 1     | 1     | 1     |
| 7   | 3   | 1     | 0     | 2     | 0     | -1    | 0     | 0     | 0     |
| 7   | 4   | 0     | -1    | -1    | 1     | 0     | 1     | 1     | 0     |
| 7   | 5   | 0     | -1    | 0     | 0     | 1     | 1     | 0     | 0     |
| 7   | 6   | -1    | -2    | -2    | 0     | 0     | 2     | 1     | 1     |
| 7   | 7   | 1     | 0     | 1     | -1    | 0     | 0     | 1     | 0     |
| 7   | 8   | -1    | -3    | -3    | 1     | 1     | 1     | 1     | 2     |
| 8   | 1   | 0     | -2    | -2    | 1     | 0     | 1     | 1     | 0     |
| 8   | 2   | -1    | -1    | -3    | 1     | 1     | 0     | 1     | 1     |
| 8   | 3   | 1     | 2     | 3     | -1    | -1    | -1    | 0     | 0     |
| 8   | 4   | 0     | 0     | 0     | 1     | 0     | -1    | 1     | 0     |
| 8   | 5   | 0     | -1    | -1    | 0     | 1     | 0     | 1     | 1     |
| 8   | 6   | -1    | -1    | -2    | 1     | 0     | 1     | 1     | 1     |
| 8   | 7   | -1    | -1    | -2    | 0     | 0     | 0     | 2     | 1     |
| 8   | 8   | 1     | 1     | 1     | 0     | -1    | -1    | 0     | 1     |

TABLE 2. Coefficients  $c_k$  such that  $L_i \cdot N_j = \sum_{k=1}^8 c_k N_k$ , for the bases defined above.



**Lemma 4.4:**  $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$ .

| $i$ | $j$ | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ |
|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|
| 1   | 1   | -1    | 1     | 1     | 0     | 0     | 0     | 0     | 0     |
| 1   | 2   | 0     | 0     | 2     | 0     | -1    | -1    | 0     | -1    |
| 1   | 3   | -2    | -2    | -4    | 0     | 1     | 1     | 0     | 1     |
| 1   | 4   | 0     | 1     | 2     | -2    | -1    | 0     | -1    | 0     |
| 1   | 5   | -2    | -1    | -2    | 1     | -1    | 0     | 0     | 1     |
| 1   | 6   | 0     | 2     | 2     | -1    | 0     | -2    | -1    | -1    |
| 1   | 7   | -2    | -2    | -2    | 1     | 1     | 1     | -1    | 0     |
| 1   | 8   | -2    | -1    | -2    | 0     | 0     | 1     | 1     | -1    |
| 2   | 1   | -2    | -3    | -2    | 0     | 1     | 1     | 0     | 1     |
| 2   | 2   | 2     | -1    | 1     | 0     | 0     | 0     | 0     | 0     |
| 2   | 3   | 2     | 3     | 3     | 0     | -2    | -2    | 0     | -2    |
| 2   | 4   | 2     | 1     | 2     | -1    | 0     | -1    | 0     | -2    |
| 2   | 5   | 4     | 4     | 7     | -2    | -3    | -2    | -1    | -2    |
| 2   | 6   | 0     | -2    | -1    | 1     | 0     | -1    | 2     | 0     |
| 2   | 7   | 2     | 1     | 2     | 0     | -1    | -2    | -1    | 0     |
| 2   | 8   | 2     | 1     | 3     | 1     | -1    | -1    | -1    | -2    |
| 3   | 1   | 0     | 1     | 0     | 0     | -1    | -1    | 0     | -1    |
| 3   | 2   | -2    | -3    | -5    | 0     | 2     | 2     | 0     | 2     |
| 3   | 3   | 2     | 1     | 1     | 0     | 0     | 0     | 0     | 0     |
| 3   | 4   | -2    | -3    | -6    | 1     | 2     | 1     | 2     | 2     |
| 3   | 5   | 0     | -2    | -3    | 0     | 1     | 2     | 1     | 0     |
| 3   | 6   | 0     | -2    | -3    | 1     | 0     | 1     | 0     | 2     |
| 3   | 7   | 2     | 3     | 2     | -2    | -1    | 0     | -1    | 0     |
| 3   | 8   | 2     | 1     | 1     | -1    | 1     | -1    | -1    | 0     |
| 4   | 1   | -2    | -1    | -2    | -1    | 1     | 0     | 1     | 0     |
| 4   | 2   | 0     | -2    | -1    | -1    | 0     | 1     | 0     | 2     |
| 4   | 3   | 4     | 4     | 5     | -1    | -2    | -1    | -2    | -2    |
| 4   | 4   | 2     | 1     | 1     | -2    | 0     | 0     | 0     | 0     |
| 4   | 5   | 2     | 2     | 3     | -2    | -2    | 1     | -1    | -1    |
| 4   | 6   | 2     | 1     | 2     | -1    | -2    | -1    | 0     | 0     |
| 4   | 7   | 4     | 5     | 7     | -3    | -2    | -2    | -3    | -1    |
| 4   | 8   | 2     | 0     | 2     | -1    | 0     | 0     | -1    | -1    |
| 5   | 1   | 0     | 1     | 2     | -1    | -2    | 0     | 0     | -1    |
| 5   | 2   | -2    | -5    | -6    | 2     | 1     | 2     | 1     | 2     |
| 5   | 3   | 2     | 3     | 2     | 0     | -1    | -2    | -1    | 0     |
| 5   | 4   | 0     | -1    | -2    | 0     | 0     | -1    | 1     | 1     |
| 5   | 5   | 2     | 1     | 1     | 0     | -2    | 0     | 0     | 0     |
| 5   | 6   | 0     | -1    | -1    | 2     | -1    | -1    | 0     | 1     |
| 5   | 7   | 0     | 0     | -1    | 0     | -1    | 0     | -1    | 2     |
| 5   | 8   | 2     | 2     | 2     | 0     | -1    | -1    | -2    | -1    |
| 6   | 1   | -2    | -2    | -2    | 1     | 0     | -1    | 1     | 1     |
| 6   | 2   | 2     | 1     | 2     | -1    | 0     | -1    | -2    | 0     |
| 6   | 3   | 2     | 3     | 2     | -1    | 0     | -1    | 0     | -2    |
| 6   | 4   | 0     | 0     | -1    | -1    | 2     | -1    | 0     | 0     |
| 6   | 5   | 2     | 2     | 2     | -2    | -1    | -1    | 0     | -1    |
| 6   | 6   | 2     | 1     | 1     | 0     | 0     | -2    | 0     | 0     |
| 6   | 7   | 2     | 4     | 3     | -1    | -1    | -2    | -2    | -1    |
| 6   | 8   | 4     | 5     | 7     | -2    | -1    | -3    | -2    | -3    |
| 7   | 1   | 0     | 2     | 2     | -1    | -1    | -1    | -2    | 0     |
| 7   | 2   | 0     | -2    | -1    | 0     | 1     | 2     | -1    | 0     |
| 7   | 3   | 0     | -2    | -3    | 2     | 1     | 0     | 1     | 0     |
| 7   | 4   | -2    | -4    | -6    | 1     | 2     | 2     | 1     | 1     |
| 7   | 5   | 2     | 1     | 2     | 0     | -1    | 0     | -1    | -2    |
| 7   | 6   | 0     | -3    | -2    | 1     | 1     | 0     | 0     | 1     |
| 7   | 7   | 2     | 1     | 1     | 0     | 0     | 0     | -2    | 0     |
| 7   | 8   | 0     | -1    | -1    | 1     | 2     | 0     | -1    | -1    |
| 8   | 1   | 0     | 1     | 2     | 0     | 0     | -1    | -1    | -2    |
| 8   | 2   | 0     | -2    | -2    | -1    | 1     | 1     | 1     | 0     |
| 8   | 3   | 0     | 0     | -2    | 1     | -1    | 1     | 1     | 0     |
| 8   | 4   | 0     | 1     | -1    | -1    | 0     | 0     | 1     | -1    |
| 8   | 5   | 0     | -1    | -1    | 0     | -1    | 1     | 2     | -1    |
| 8   | 6   | -2    | -4    | -6    | 2     | 1     | 1     | 2     | 1     |
| 8   | 7   | 2     | 2     | 2     | -1    | -2    | 0     | -1    | -1    |
| 8   | 8   | 2     | 1     | 1     | 0     | 0     | 0     | 0     | -2    |

TABLE 3. Coefficients  $c_k$  such that  $M_i \cdot M_j = \sum_{k=1}^8 c_k M_k$ , for the bases defined above.

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